UNIT 4 · D01

Procedural programming

UNIT 4 · D01

Object-oriented programming

UNIT 4 · D01

Class

UNIT 4 · D01

Object

UNIT 4 · D01

Encapsulation

UNIT 4 · D01

Inheritance

UNIT 4 · D02

IDE

UNIT 4 · D02

IntelliSense / auto-complete

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Programs built from self-contained objects that combine data (properties) and methods. Object-oriented programming	Step-by-step instructions executed in sequence, organised into procedures and functions. Procedural programming
An instance created from a class with its own data. Object	A blueprint defining the properties and methods of objects. Class
A class taking on the properties and methods of another class. Inheritance	Keeping an object's data inside it, accessed only through its methods. Encapsulation
Editor feature that suggests or completes code as it is typed. IntelliSense / auto-complete	Integrated development environment: editor, translator, debugger and tools in one application. IDE

UNIT 4 · D02

Translator

UNIT 4 · D02

Machine code

UNIT 4 · D02

Compiler

UNIT 4 · D02

Interpreter

UNIT 4 · D03

Denary

UNIT 4 · D03

Binary

UNIT 4 · D04

Hexadecimal

UNIT 4 · D05

Carry

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Binary instructions the processor executes directly. Machine code	Software that converts source code into machine code: a compiler or interpreter. Translator
Translates and executes source code one line at a time. Interpreter	Translates the whole program into machine code before running, producing a separate executable. Compiler
Base-2 number system using 0 and 1. Binary	Base-10 number system. Denary
The digit passed to the next column when a binary column total exceeds 1. Carry	Base-16 number system (0–9, A–F); a shorthand for binary. Hexadecimal

UNIT 4 · D05

Overflow

UNIT 4 · D06

ASCII

UNIT 4 · D06

Unicode

UNIT 4 · D07

Boolean

UNIT 4 · D07

Float / real

UNIT 4 · D08

Truth table

UNIT 4 · D08

AND / OR / NOT

UNIT 4 · D09

Top-down design

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American Standard Code for Information Interchange: 7-bit (128) or 8-bit (256) character set, mainly English. ASCII	When a result is too large for the available bits, so the extra bit is lost and the result is wrong. Overflow
Data type with only two values, True or False. Boolean	Character set covering almost all languages and emoji, using up to 32 bits per character; includes ASCII. Unicode
Table showing the output for every combination of inputs. Truth table	Data type for decimal numbers. Float / real
Breaking the main task into smaller sub-tasks. Top-down design	AND is true only if both inputs are true; OR is true if at least one is; NOT reverses the input. AND / OR / NOT

UNIT 4 · D09

Abstraction

UNIT 4 · D09

Decomposition

UNIT 4 · D10

Algorithm

UNIT 4 · D10

Pseudocode

UNIT 4 · D10

Flowchart

UNIT 4 · D11

Bubble sort

UNIT 4 · D11

Insertion sort

UNIT 4 · D12

Linear search

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Breaking a complex problem into smaller, more manageable sub-problems. Decomposition	Removing details of a problem that are not needed for the solution. Abstraction
Language-independent, English-like instructions describing an algorithm. Pseudocode	A step-by-step plan for solving a problem. Algorithm
Compares adjacent items and swaps them if out of order, repeating passes until no swaps occur. Bubble sort	Diagram of an algorithm using standard symbols for start/stop, input/output, process and decision. Flowchart
Checks each item in turn until the target is found or the list ends. Linear search	Builds a sorted list by inserting each item into its correct position. Insertion sort

UNIT 4 · D12

Binary search

UNIT 4 · D13

Dry run

UNIT 4 · D13

Trace table

UNIT 4 · D13

Desk checking

UNIT 4 · D14

Variable

UNIT 4 · D14

Constant

UNIT 4 · D15

Sequence

UNIT 4 · D15

Selection

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Working through code by hand, line by line, recording variable values. Dry run	Repeatedly checks the middle of a sorted list and discards half; needs sorted data. Binary search
Reviewing code by hand to find errors before running it. Desk checking	Table recording the values of variables at each step of a dry run. Trace table
A named value that stays the same while the program runs. Constant	A named memory location whose value can change while the program runs. Variable
A condition decides which statements run (IF statements). Selection	Statements executed one after another in order. Sequence

UNIT 4 · D16

Iteration

UNIT 4 · D16

Infinite loop

UNIT 4 · D17

Array

UNIT 4 · D17

Index

UNIT 4 · D18

Concatenation

UNIT 4 · D18

String function

UNIT 4 · D19

Function

UNIT 4 · D19

Procedure

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A loop whose condition never becomes false, so it never ends. Infinite loop	Repeating a block of code (loops). Iteration
The position number of an element in an array (starting at 0). Index	A data structure holding values of one data type under one name, accessed by index. Array
A function that manipulates text, e.g. length, split, find, substring. String function	Joining strings together. Concatenation
A named subroutine that performs a task without returning a value. Procedure	A named subroutine that performs a task and returns a value. Function

UNIT 4 · D19

Parameter

UNIT 4 · D20

File handling

UNIT 4 · D21

Syntax error

UNIT 4 · D21

Execution / run-time error

UNIT 4 · D21

Logic error

UNIT 4 · D21

Error trapping

UNIT 4 · D22

White box testing

UNIT 4 · D22

Black box testing

Backs · print double-sided, flip on long edge

Opening, reading, writing, appending and closing files to store data permanently. File handling	A value passed into a subroutine. Parameter
An error that occurs while the program runs and causes it to crash, e.g. divide by zero. Execution / run-time error	Breaking the grammar rules of the language; found by the translator before running. Syntax error
Code that detects a run-time error and handles it instead of crashing, e.g. try/except. Error trapping	The program runs but gives incorrect results because the reasoning is wrong. Logic error
Testing inputs and outputs against requirements without seeing the code. Black box testing	Testing the internal code, paths and logic, usually by the programmer. White box testing

UNIT 4 · D22

Unit testing

UNIT 4 · D22

Integration testing

UNIT 4 · D22

System testing

UNIT 4 · D22

Test plan

UNIT 4 · D22

Valid / normal data

UNIT 4 · D22

Extreme / boundary data

UNIT 4 · D22

Invalid / erroneous data

UNIT 4 · D22

Null data

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Testing that separately built units work together. Integration testing	Testing a single module, function or method on its own. Unit testing
Table listing each test, its purpose, test data, expected outcome, actual outcome and action. Test plan	Testing the complete, integrated system against its specification. System testing
Values at the edges of the accepted range. Extreme / boundary data	Typical data that should be accepted. Valid / normal data
No data entered. Null data	Data that should be rejected. Invalid / erroneous data

UNIT 4 · D23

Evaluation

UNIT 4 · D23

User requirements

UNIT 4 · D23

Robustness

UNIT 4 · D23

Stakeholders

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The agreed list of what the system must do, used to judge the solution. User requirements	Judging the solution against the user requirements and design. Evaluation
People with an interest in the system: client, end users, developers. Stakeholders	The ability to handle invalid, unexpected and high volumes of data without failing. Robustness